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Also we know,

$$d\vec{r} = dr\,\hat{r} \tag{6.24}$$

$$\Rightarrow dW = \frac{Gm_1m_2}{r^2}\hat{r}.(dr\,\hat{r}) \tag{6.25}$$

$$\hat{r}.\hat{r} = 1 \left(since\ both\ are\ unit\ vectors\right)$$

$$\therefore \quad dW = \frac{Gm_1m_2}{r^2}dr \tag{6.26}$$

Thus the total work done for displacing the particle from r' to r is

$$W = \int_{r'}^{r} dW = \int_{r'}^{r} \frac{Gm_1m_2}{r^2}dr \tag{6.27}$$

$$W = -\left(\frac{Gm_1m_2}{r}\right)_{r'}^{r}$$

$$W = -\frac{Gm_1m_2}{r} + \frac{Gm_1m_2}{r'} \tag{6.28}$$

$$W = U(r) - U(r')$$

where $U(r) = \dfrac{-Gm_1m_2}{r}$

This work done W gives the gravitational potential energy difference of the system of masses m_1 and m_2 when the separation between them are r and r' respectively.

Case 1: If $r < r'$

Since gravitational force is attractive, m_2 is attracted by m_1.Then m_2 can move from r to r' without any external work (Figure 6.13). Here work is done by the system spending its internal energy and hence the work done is said to be negative.

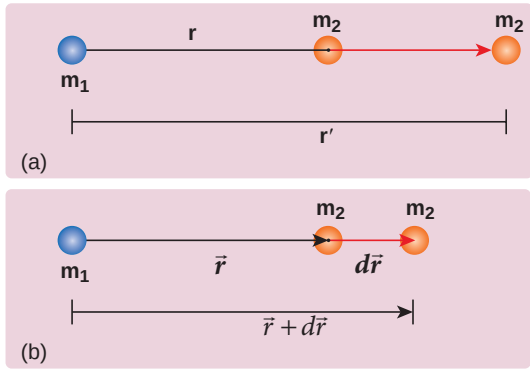


Figure 6.13 Cases for calculation of work done by gravity

Case 2: If $r > r'$

Work has to be done against gravity to move the object from r' to r. Therefore work is done on the body by external force and hence work done is positive.

It is to be noted that only potential energy difference has physical significance. Now gravitational potential energy can be discussed by choosing one point as the reference point.

Let us choose $r' = \infty$. Then the second term in the equation (6.28) becomes zero.

$$W = -\frac{Gm_1m_2}{r} + 0 \tag{6.29}$$

Now we can define gravitational potential energy of a system of two masses m_1 and m_2 separated by a distance r as the amount of work done to bring the mass m_2 from infinity to a distance r assuming m_1 to be fixed in its position and is written as $U(r) = -\dfrac{Gm_1m_2}{r}$. It is to be noted that the gravitational potential energy of the system consisting of two masses m_1 and m_2 separated by a distance r, is the gravitational potential energy difference of the system when the masses are